FOR CLOSED IMMERSIONS OF LOCALLY NOETHERIAN FORMAL SPACES, RIG-SMOOTH EQUALS RIG-ETALE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof first identifies the ring maps corresponding to etale-local pieces of a closed immersion, then verifies the two implications through the description of NL^wedge as J/J^2 in degree -1, closing with the observation that a power of an ideal of definition is again an ideal of definition

Editorial note. The label lemma-closed-immersion-rig-smooth names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- closed immersions of locally countably indexed formal algebraic spaces correspond to surjective morphisms B -> A in WAdm^Noeth (Formal Spaces, Lemma formal-spaces-lemma-closed-immersion-into-countably-indexed)
- rig-etale morphisms are rig-smooth (lemma-rig-etale-rig-smooth-morphism of the same chapter)
- the definition of rig-etale morphisms of formal algebraic spaces via the ring maps of WAdm^Noeth (lemma-rig-etale-morphisms and definition-rig-etale-continuous-homomorphism of the same chapter)
- the completed naive cotangent complex of a quotient B -> A = B/J is the complex with J/J^2 in degree -1 (definition-rig-etale-homomorphism of the same chapter)
- rig-smooth ring maps and the criterion that annihilation of J/J^2 by a power of I follows from rig-smoothness (definition-rig-smooth and lemma-equivalent-with-artin-smooth of the same chapter)

Source reference. Original reference 0GCV: sources/stacks-restricted/pinned-original-source.tex, source lines 4776–4776. Prerequisite pointer invoked at proof line(s) 5211; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAY: sources/stacks-restricted/pinned-original-source.tex, source lines 1794–1794. Prerequisite pointer invoked at proof line(s) 5216; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0GCP: sources/stacks-restricted/pinned-original-source.tex, source lines 4622–4622. Prerequisite pointer invoked at proof line(s) 5220; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AQI: sources/stacks-chow/tags.txt, source lines 13511–13511. Prerequisite pointer invoked at proof line(s) 5200; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAJ: sources/stacks-restricted/pinned-original-source.tex, source lines 843–843. Prerequisite pointer invoked at proof line(s) 5225; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GCZ: sources/stacks-restricted/pinned-original-source.tex, source lines 5047–5047. Prerequisite pointer invoked at proof line(s) 5208; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GD0: sources/stacks-restricted/pinned-original-source.tex, source lines 5114–5114. Prerequisite pointer invoked at proof line(s) 5204; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of formal algebraic spaces. Assume that X and Y are locally Noetherian and f is a closed immersion. The following are equivalent*

1. *f is rig-smooth,*
2. *f is rig-étale,*
3. *for every affine formal algebraic space V and every morphism $V \to Y$ which is representable by algebraic spaces and étale the morphism $X \times_Y V \to V$ corresponds to a surjective morphism $B \to A$ in $WAdm^{Noeth}$ whose kernel J has the following property: $I(J/J^2) = 0$ for some ideal of definition I of B.*

Proof. Let us observe that given V and $V \to Y$ as in (2) without any further assumption on f we see that the morphism $X \times_Y V \to V$ corresponds to a surjective morphism $B \to A$ in $WAdm^{Noeth}$ by Formal Spaces, Lemma 0AQI.

We have (2) $\Rightarrow$ (1) by Lemma 0GD0.

Proof of (3) $\Rightarrow$ (2). Assume (3). By Lemma 0GCZ it suffices to show that the ring maps $B \to A$ occurring in (3) are rig-étale in the sense of Definition 0GCV. Let I be as in (3). The naive cotangent complex $NL^\wedge_{A/B}$ of A over (B, I) is the complex of A-modules given by putting J/J^2 in degree -1. Hence A is rig-étale over (B, I) by Definition 0GAY.

Assume (1) and let V and $B \to A$ be as in (3). By Definition 0GCP we see that $B \to A$ is rig-smooth. Choose any ideal of definition $I \subset B$. Then A is rig-smooth over (B, I). As above the complex $NL^\wedge_{A/B}$ is given by putting J/J^2 in degree -1. Hence by Lemma 0GAJ we see that J/J^2 is annihilated by a power I^n for some

$n \geq 1$. Since B is adic, we see that I^n is an ideal of definition of B and the proof is complete. $\square$